

Invariant Concept Extraction from Controlled Contrasts

Anonymous ACL submission

Abstract

Placeholder. We study controlled contrastive learning for separating invariant and varying information in frozen representations. The abstract will be completed once the empirical setting and results are finalized.

1 Introduction

Placeholder. Motivation, contributions, and empirical context will be added here.

2 Related Work

Placeholder. Relevant work will be added here.

3 Method

Placeholder. The implementation and data-construction details will be added here.

4 Theoretical Analysis

4.1 Generative setting and controlled pairs

Assume that an observation is generated by two latent blocks,

$$X = g(C, S), \quad H = F(X), \quad (1)$$

where C denotes information intended to remain invariant under a chosen transformation relation, S denotes information allowed to vary, and F is a frozen representation model. We learn two maps,

$$Z_C = f_C(H), \quad Z_S = f_S(H). \quad (2)$$

We do not assume $C \perp S$. The objective is relative specialization: Z_C should respond more strongly to changes in C than to changes in S , and conversely for Z_S .

For an anchor $x_a = g(c_a, s_a)$, construct a C -positive $x_p = g(c_p, s_p)$ and a C -negative $x_n =$

$g(c_n, s_n)$. Rather than assuming perfect interventions, we assume bounded fidelity:

$$d_C(c_a, c_p) \leq \eta_C, \quad d_S(s_a, s_p) \geq \kappa_S, \quad (3)$$

$$d_S(s_a, s_n) \leq \eta_S, \quad d_C(c_a, c_n) \geq \kappa_C. \quad (4)$$

The ideal controlled setting has $\eta_C = \eta_S = 0$.

4.2 Matched contrastive objective

For the invariant branch, let

$$u^+ = \text{sim}(f_C(h_a), f_C(h_p)), \quad (5)$$

$$u^- = \text{sim}(f_C(h_a), f_C(h_n)). \quad (6)$$

With temperature $\tau > 0$, the matched objective is

$$\mathcal{L}_C = -\log \frac{\exp(u^+/\tau)}{\exp(u^+/\tau) + \exp(u^-/\tau)} \quad (7)$$

$$= \log \left(1 + \exp \left(\frac{u^- - u^+}{\tau} \right) \right). \quad (8)$$

For Z_S , we exchange the controlled contrasts: approximately equal S and different C is positive, while approximately equal C and different S is negative. The partition objective is $\mathcal{L}_{\text{part}} = \mathcal{L}_C + \mathcal{L}_S$. It requires neither reconstruction nor exact statistical independence.

Unrestricted in-batch InfoNCE is unsuitable for this purpose: it pushes down the similarity of every non-positive example, even though $c_i \neq c_j$ does not imply that two observations share no invariant information. They may share an entity, relation, topic, or another component of C . A matched negative instead controls S , so the contrast cannot be solved merely through a difference in S .

Proposition 1 (Relative margin). *If $\mathcal{L}_C \leq \epsilon$ for a training triplet, then*

$$u^+ - u^- \geq \tau \log \frac{1}{\exp(\epsilon) - 1}. \quad (9)$$

In particular, when $\epsilon < \log 2$, $\text{sim}(Z_C^a, Z_C^p) > \text{sim}(Z_C^a, Z_C^n)$.

Proof. Equation (8) gives $\exp((u^- - u^+)/\tau) \leq \exp(\epsilon) - 1$. Taking logarithms and rearranging proves the claim. $\square$

4.3 Regularity assumptions

Write $z_C(c, s)$ for the invariant-branch representation. In addition to bounded fidelity, assume the following.

Assumption 1 (Observed alignment and separation). *For observed positive pairs, $d_Z(z_C(c_a, s_a), z_C(c_p, s_p)) \leq \delta_{\text{obs}}$. For observed matched negatives, $d_Z(z_C(c_a, s_a), z_C(c_n, s_n)) \geq \gamma_{\text{obs}}$.*

Assumption 2 (Local regularity). *The representation is locally Lipschitz in each block:*

$$d_Z(z_C(c, s), z_C(c', s)) \leq L_C d_C(c, c'), \quad (10)$$

$$d_Z(z_C(c, s), z_C(c, s')) \leq L_S d_S(s, s'). \quad (11)$$

Assumption 3 (Nuisance coverage). *For every unseen nuisance realization s^* , there is a covered training realization $\tilde{s}$ with $d_S(s^*, \tilde{s}) \leq \epsilon$. The observed alignment and separation conditions hold for the corresponding covered controlled pairs.*

Pair imperfection yields the effective bounds

$$\delta = \delta_{\text{obs}} + L_C \eta_C, \quad \gamma = \gamma_{\text{obs}} - L_S \eta_S. \quad (12)$$

Theorem 1 (Unseen- S stability). *Let s^* and s^{**} be unseen nuisance realizations with covered training realizations within ϵ . For fixed c ,*

$$d_Z(z_C(c, s^*), z_C(c, s^{**})) \leq \delta_{\text{obs}} + L_C \eta_C + 2L_S \epsilon. \quad (13)$$

Proof. Apply the triangle inequality through the two covered nuisance realizations, then Assumption 2 to each unseen-to-covered segment and Equation (12) to the middle segment. $\square$

Theorem 2 (Unseen- C separation). *For $c_i \neq c_j$ under an unseen nuisance s^* ,*

$$d_Z(z_C(c_i, s^*), z_C(c_j, s^*)) \geq \gamma_{\text{obs}} - L_S \eta_S - 2L_S \epsilon. \quad (14)$$

Proof. Use the reverse triangle inequality through a covered nuisance realization, then Assumption 2 and Equation (12). $\square$

Corollary 1 (Sufficient operating condition). *Same- C observations remain closer than different- C observations whenever*

$$\gamma_{\text{obs}} > \delta_{\text{obs}} + L_C \eta_C + L_S \eta_S + 4L_S \epsilon. \quad (15)$$

This condition is sufficient, not necessary.

4.4 Dependence imposes unavoidable leakage

Theorem 3 (Leakage lower bound). *For discrete C and any learned representation Z_C ,*

$$I(Z_C; S) \geq I(C; S) - H(C | Z_C). \quad (16)$$

Proof. The chain rule gives $I(C, Z_C; S) = I(C; S) + I(Z_C; S | C) = I(Z_C; S) + I(C; S | Z_C)$. Non-negativity of $I(Z_C; S | C)$ and $I(C; S | Z_C) \leq H(C | Z_C)$ establish the result. $\square$

If C is decoded from Z_C with error probability p_e , Fano's inequality yields

$$I(Z_C; S) \geq I(C; S) - h(p_e) - p_e \log(|C| - 1). \quad (17)$$

Thus, when C and S are dependent, faithful preservation of C entails a nonzero leakage floor. The appropriate goal is to remove S information beyond that intrinsically coupled to C , rather than to require $I(Z_C; S) = 0$.

4.5 Propagation to sparse autoencoders

Let $A = \phi(Z_C)$ be an SAE encoding. For a Top- k SAE, suppose locally that $\|\phi(z) - \phi(z')\| \leq L_{\text{SAE}} \|z - z'\|$ while its selected support is fixed. Theorem 1 then gives

$$\begin{aligned} & \|\phi(z_C(c, s^*)) \\ & \quad - \phi(z_C(c, s^{**}))\| \\ & \leq L_{\text{SAE}} (\delta_{\text{obs}} + L_C \eta_C + 2L_S \epsilon). \end{aligned} \quad (18)$$

Moreover, if ordered SAE preactivations have Top- k support margin $\rho = q_{(k)} - q_{(k+1)}$, the support is unchanged whenever $\|W \Delta z_C\|_\infty < \rho/2$. A sufficient condition is

$$\|W\|_{\infty \leftarrow 2} (\delta_{\text{obs}} + L_C \eta_C + 2L_S \epsilon) < \rho/2. \quad (19)$$

Scope. The analysis establishes controlled relative preference, bounded nuisance drift under the stated assumptions, and downstream feature stability under a support-margin condition. It does not prove $Z_C = C$, $Z_S = S$, $Z_C \perp S$, globally identifiable factors, human-interpretable concepts, or generalization to arbitrarily distant nuisance variation.

5 Experiments

Placeholder. Experimental design and results will be added here.

6 Conclusion

Placeholder. The conclusion will be added here.